

Question Paper

Exam Date & Time: 27-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH EXAMINATIONS - NOVEMBER 2025

Advanced Computer Architecture [ECE_5421]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Suppose you are designing a 32-bit RISC ISA with 16 general-purpose registers. You can choose between: (i) 3-operand instruction format, and (ii) 2-operand instruction format. (5)

A) Compare their impact on:

- Instruction encoding length
- Datapath complexity
- Register file read/write ports

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) A datapath includes the following stage delays: (5)

- Instruction Memory: 250 ps
- Register File Read: 120 ps
- ALU: 180 ps
- Data Memory: 250 ps
- Register Write: 100 ps

(i) Compute the critical path delay for a single-cycle CPU.

(ii) Discuss how splitting the datapath into multiple cycles or pipeline stages would improve performance.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2) Differentiate between volatile and non-volatile memories and Explain the difference between ROM, PROM, EPROM, and EEPROM. (4)

A)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- B) Explain how cache coherence is maintained in multi-core systems. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- C) A processor has a 5-stage pipeline: IF, ID, EX, MEM, WB. (4)
Given the instruction sequence below, identify any hazards and describe how they can be resolved (with or without stalls):

1. LW R1, 0(R2)
2. ADD R3, R1, R4
3. SUB R5, R3, R6

4. SW R5, 8(R2)

Identify all data hazards (RAW, WAR, WAW) and Indicate which can be resolved by forwarding and which require stalling.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) What are the main limitations of a single-cycle CPU in terms of clock period and performance? (2)

A) Course Outcome: "CO3" / Cognitive Level: "Understand"

- B) Discuss trade-offs in area, power, speed, and control complexity between single-cycle and multi-cycle CPUs when implemented in VLSI. Include physical design considerations like clock distribution and critical path reduction. (5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) Suppose a single-cycle CPU's critical path runs through:
Instruction Memory → Register File → ALU → Data Memory → Write Back
Propose one architectural modification to reduce the critical path delay. Justify how each change impacts CPI and hardware complexity. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) Compare the hardware resource utilization between single-cycle and multi-cycle designs. (2)

A) Course Outcome: "CO4" / Cognitive Level: "Understand"

- B) A single-cycle processor has the following component delays: (4)

- Instruction Memory = 200 ps
- Register File Read = 100 ps
- ALU = 150 ps
- Data Memory = 250 ps
- Register Write = 100 ps

(i) Determine the clock period for a single-cycle CPU.

(ii) Suggest how a multi-cycle design can reduce this clock period, and estimate the improvement.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Explain how registers, ALUs are shared among cycles in a multi-cycle CPU to reduce hardware cost. Provide an example showing which hardware blocks are reused. (4)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) Compare direct-mapped, fully associative, and set-associative cache organizations. (3)

A) Course Outcome: "CO5" / Cognitive Level: "Understand"

- B) Define memory hierarchy in computer architecture. Explain why it is essential for bridging the processor-memory speed gap. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- C) What are the major steps in interrupt handling in a CPU? (4)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025-PG ECE_5421:Advanced Computer Architecture

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG ECE_5421:Advanced Computer Architecture

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	8	24(48%)
3	Application	0	0(0%)
4	Analysis	6	26(52%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		14	50 (100%)